

MERRILL, WI, USA

WMO: 726449

Lat: 45.199N

Lon: 89.711W

Elev: 1317

StdP: 14.01

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04891

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-10.6	-24.1	1.5	-15.0	-18.2	2.1	-8.7	19.2	18.5	17.7	19.2	1.4	290	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	85.6	71.6	82.2	69.6	80.6	68.1	74.5	81.8	72.6	79.5	70.8	77.2	8.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	125.7	78.6	70.3	117.2	76.6	68.6	110.2	74.7	38.8	81.3	37.1	79.1	35.4	77.2	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	16.5	14.2	DB	-25.1	90.3	3.3	3.2	-27.5	92.6	-29.4	94.4	-31.3	96.2	-33.7	98.5	
			WB	-25.5	77.4	3.2	2.6	-27.8	79.3	-29.7	80.8	-31.5	82.3	-33.8	84.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	14.2	15.5	28.0	41.0	53.2	62.4	66.9	64.2	57.3	44.5	32.1	19.6
	DBStd	20.91	12.68	11.82	12.55	9.57	8.61	6.43	5.80	5.96	8.11	8.63	9.67	11.20
	HDD50	4856	1111	966	690	298	63	2	0	1	22	221	540	942
	HDD65	8750	1576	1386	1147	720	379	121	47	86	255	639	987	1407
	CDD50	1833	0	0	8	29	161	374	524	442	243	49	3	0
	CDD65	253	0	0	0	1	12	44	106	62	25	3	0	0
	CDH74	2869	0	0	1	26	199	502	1193	657	271	20	0	0
	CDH80	590	0	0	0	3	42	88	292	117	47	1	0	0
Wind	WSAvg	5.2	5.2	5.3	5.6	6.4	6.1	5.0	4.3	4.0	4.5	5.5	5.3	5.2
Precipitation	PrecAvg	32.7	1.1	1.0	1.9	2.8	3.5	3.8	3.9	3.9	4.4	2.6	2.1	1.5
	PrecMax	41.4	3.3	2.5	4.3	4.4	7.0	8.3	9.4	8.6	8.7	5.1	4.7	3.5
	PrecMin	20.4	0.0	0.0	0.2	0.8	1.1	1.1	1.5	1.1	0.6	0.2	0.2	0.4
	PrecStd	5.2	0.7	0.8	1.2	1.0	1.6	2.1	1.9	1.9	2.2	1.3	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	44.9	70.1	78.6	85.0	88.0	90.5	86.9	85.6	76.8	62.9	46.6
		MCWB	37.3	38.0	59.9	60.7	67.7	74.3	76.2	71.6	71.6	62.8	53.3	41.6
	2%	DB	37.1	38.9	59.0	70.4	79.4	81.9	86.3	82.9	80.9	71.7	55.4	40.7
		MCWB	34.7	34.6	50.1	55.3	64.7	69.2	72.6	70.6	68.3	62.7	49.7	38.4
	5%	DB	34.7	35.8	52.1	63.8	74.9	79.5	82.4	80.7	77.1	65.5	51.7	36.7
		MCWB	33.2	32.6	46.0	50.0	60.8	66.9	70.6	68.3	65.9	57.7	45.7	34.9
	10%	DB	32.0	32.2	45.3	59.1	70.4	76.6	80.8	77.8	73.1	60.2	46.4	34.4
		MCWB	30.9	30.3	39.7	47.3	58.3	65.0	69.0	67.0	64.5	53.1	42.5	33.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.6	39.2	60.3	62.8	71.3	76.1	78.1	75.3	73.7	70.0	55.7	45.0
		MCDB	40.6	43.6	69.6	73.4	80.5	84.7	86.3	82.0	81.0	73.1	59.4	45.5
	2%	WB	35.4	35.8	53.0	57.4	67.3	72.3	75.1	73.2	71.1	62.8	51.2	38.2
		MCDB	37.2	38.8	56.7	69.6	75.2	79.5	82.8	79.2	77.1	68.0	55.9	39.7
	5%	WB	33.0	32.4	45.1	53.1	64.2	69.9	73.2	71.1	69.0	58.7	46.3	35.4
		MCDB	34.6	34.6	51.4	60.7	71.4	76.3	80.2	77.3	74.4	64.7	50.0	36.6
	10%	WB	30.6	30.4	40.2	48.6	60.6	67.5	71.1	69.2	66.2	54.2	42.6	33.0
		MCDB	31.6	31.9	45.1	57.4	67.8	73.9	78.0	74.9	71.5	59.0	46.8	34.2
Mean Daily Temperature Range	5% DB	MDBR	17.0	20.2	21.0	22.0	24.1	22.7	23.8	23.4	23.7	19.6	15.9	14.8
		MCDBR	16.3	19.8	26.5	32.9	29.5	25.7	26.0	26.2	27.3	28.4	24.9	16.5
		MCWBR	14.0	15.8	17.5	18.9	15.3	14.4	13.9	14.3	15.9	18.0	17.8	14.6
	5% WB	MCDBR	15.2	17.6	24.1	28.9	24.0	22.2	22.6	21.4	21.3	24.8	21.8	13.9
		MCWBR	13.3	14.6	16.9	18.7	13.9	13.8	13.4	12.9	13.8	17.7	17.1	12.7
Clear-Sky Solar Irradiance	taub	0.245	0.274	0.313	0.343	0.373	0.372	0.369	0.371	0.345	0.311	0.283	0.253	
	taud	2.382	2.298	2.353	2.392	2.346	2.368	2.393	2.391	2.464	2.547	2.529	2.443	
	Ebn at Noon	282	290	292	291	284	284	283	279	278	275	266	267	
	Edh at Noon	23	30	34	36	38	38	36	35	30	24	20	20	
All-Sky Solar Radiation	RadAvg	511	864	1234	1449	1707	1860	1920	1628	1278	799	488	362	
	RadStd	54	83	125	144	138	148	98	108	88	84	50	41	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page